

Topic to discuss

→ Proof of Lagrange's Interpolation

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Lagrange's Interpolation Formula

$$\begin{aligned} y(x) = & \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} y_1 \\ & + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} y_3 \\ & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} y_4 \end{aligned}$$

In general,

$$\begin{aligned} f(x) = & \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 \\ & + \dots + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n \end{aligned}$$

Proof: Let $y=f(x)$ be a function which takes the values $(x_0, y_0), (x_1, y_1) \dots (x_n, y_n)$.

Since there are $n+1$ pair of values of x and y , we can represent $f(x)$ by polynomial in x of degree n .

Let this polynomial be of the form,

$$\begin{aligned} y = f(x) = & a_0 (x-x_1)(x-x_2) \dots (x-x_n) + \\ & a_1 (x-x_0)(x-x_2) \dots (x-x_n) + \\ & a_2 (x-x_0)(x-x_1)(x-x_3) \dots (x-x_n) + \\ & \vdots \\ & a_n (x-x_0)(x-x_1) \dots (x-x_{n-1}) \end{aligned} \quad \text{--- (1)}$$

Putting $x = x_0$, $y = y_0$ in equation (1), we get

$$y_0 = a_0 (x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)$$

$$\Rightarrow a_0 = \frac{y_0}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)}$$

Similarly putting $x = x_1$, $y = y_1$ in equation (1), we get,

$$y_1 = a_1 (x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)$$

$$\Rightarrow a_1 = \frac{y_1}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)}$$

Proceeding the same way, we will get $a_2, a_3, \dots, a_n$

Substituting the values of $a_0, a_1, \dots, a_n$ in equation-(1)
we get,

$$y = f(x) = a_0 (x-x_1)(x-x_2)\dots(x-x_n) + \\ a_1 (x-x_0)(x-x_2)\dots(x-x_n) + \\ a_2 (x-x_0)(x-x_1)(x-x_3)\dots(x-x_n) + \\ \vdots \\ a_n (x-x_0)(x-x_1)\dots(x-x_{n-1})$$

$$\Rightarrow f(x) = \frac{y_0}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} (x-x_1)(x-x_2)\dots(x-x_n) + \\ + \frac{y_1}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} (x-x_0)(x-x_2)\dots(x-x_n) + \\ + \dots \\ + \frac{y_n}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} (x-x_0)(x-x_1)\dots(x-x_{n-1})$$

This is called Lagrange's Interpolation formula.

(Proved)

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